

Synergistic Activities — Jonathan Samir Matthis

1. **Development and open distribution of accessible research tooling.** Created and maintains FreeMoCap, a free, open-source markerless motion capture platform, and deliberately architected it as a suite of modular, independently documented repositories (camera synchronization, pose estimation, data processing, visualization) so that researchers can understand, deploy, and extend individual pipeline stages in their own projects rather than adopting an opaque monolith.
2. **Technical training and community support at scale.** Provides extensive informal training and technical support to a community of several thousand students, early-career researchers, and professionals — including guidance on conducting and publishing their work in independent, publicly visible ways — through community channels, written documentation, and regular live tutorial streams.
3. **Public, reproducible research-software development.** Conducts the entire development process “in public”: all work is hosted openly on GitHub with intentional documentation of both the technical pipeline and the organizational processes behind it, explicitly to make it easier for other groups to follow the same open, reproducible development model.
4. **Contributions to the broader open-source and research community.** Regularly contributes code review, pull requests, and technical guidance to third-party open-source projects, and provides public technical feedback to researchers presenting motion-capture and computer-vision methods work outside traditional review venues.
5. **Public science communication.** Produces publicly available video content (YouTube tutorials and Twitch live streams) on motion capture methods and open-source scientific software, and has discussed accessible research tooling and open science in podcast appearances.